

An image dataset for classification of vegetable

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Abstract

Accurate vegetable recognition is very essential in quality control and inventory management amidst the increasing demand for smart agriculture and automated food processing systems. In this effort, we have shown the development of an image-based classification system for six widely consumed vegetables: cabbage, lettuce, mustard greens, radish leaves, red amaranth, and spinach. Image data on vegetables were acquired and preprocessed by the system through corrupted image detection and removal from the dataset, resizing all images to a fixed pixel size of 224×224 , converting all color images to grayscale, normalizing the data, and removing irrelevant features. Rotation, flipping, shearing, and scaling are involved in the data augmentation process to further enhance the model's generalization capability. Various deep learning models were explored: Convolutional Neural Networks, DenseNet201, ResNet50, and Recurrent Neural Networks, for which transfer learning was utilized from pre-trained architectures. An 80:20 split was used for training and validation of the models, along with mechanisms for early stopping and checkpointing to avoid overfitting. The models were evaluated in terms of accuracy and loss curves, confusion matrices, classification reports, and ROC curves on multi-class metrics.

1 Introduction

Recognizing and classifying vegetables has become very important in today's era of smart agriculture and automated food processing. Traditional methods using manual inspection are time-consuming and labor-intensive, inviting human error, particularly in big farming, retail, and supply chains. Image-based classification uses artificial intelligence and deep learning to respond effectively to such challenges, enabling faster, accurate, and consistent recognition of vegetable varieties.

The classification in this research covers six types of common vegetables: cabbage, lettuce, mustard greens, radish leaves, red amaranth, and spinach. The proposed system takes advantage of the vegetable image dataset that undergoes exhaustive pre-processing, which includes corrupted image detection, resizing, grayscale conversion, normalization, and removal of unnecessary features. Data augmentation, such as rotation, flipping, shearing, and scaling, is also applied to improve model generalization and robustness.

To extract complex hierarchical features from the images, deep learning models like CNN, DenseNet201, ResNet50, and RNN are employed. Considering the limited dataset size, pre-trained architecture transfer learning is used to improve the performance of the models. The models are evaluated in terms of their accuracy, loss curve, confusion matrices, classification reports, and ROC curves to have a thorough understanding of the models.

2 Problem Statement

Most of the vegetable identification and classification tasks in agriculture, retail, and food processing industries are done manually, which can be slow, labor-intensive, and error-prone. Incorrect classification could lead to inefficiencies in inventory management, supply chain operations, and quality control, ultimately contributing to financial losses and lower consumer satisfaction. Classic systems also lack scalability, making it difficult to efficiently process large volumes of produce.

In particular, the challenge is to develop an automated system for the recognition and classification of several vegetable varieties from images, considering changes in lighting conditions, orientations, sizes, and backgrounds. The system must be able to work with limited datasets and still provide high accuracy and robustness. Besides, some preprocessing steps, such as corrupted image detection, resizing, normalizing, and data augmentation, are absolutely necessary to enhance model performance and generalization.

This project addresses these challenges by implementing deep learning-based image classification models that include CNN, DenseNet201, ResNet50, and RNN architectures, with the application of transfer learning where suitable. The goal is to develop an effective, reliable, and scalable solution for vegetable image recognition that reduces human effort, minimizes errors, and enhances productivity in various agricultural and food supply chain applications.

3 Related Work

- Automatic classification of images of plants and vegetables has rapidly improved thanks to the use of deep learning, especially with convolutional neural networks. The extensive review by Khanna et al., 2024, presents the state-of-the-art in deep learning applied to plant disease and species classification, pointing out the extremely high accuracy reached in datasets such as PlantVillage and other plant-leaf collections by models such as ResNet-50, EfficientNet, and Vision Transformers.
- Beyond standard CNNs, architecture innovations continue. Goyal et al. (2024) proposed the enhanced CNN (EnConv) for wheat disease classification, which surpassed not only the VGG-16, but also the ResNet-50 in terms of accuracy and precision.
- More compact CNN architectures have also been explored: Wagle, Harikrishnan, Ali & Faseehuddin (2021) designed a lightweight CNN for classifying nine species of plants, showing that efficient models can still generalize well for classification based on leaves.
- Wen & He (2024) proposed an improved EfficientNet-CBAM model for vegetable quality grading, using a dataset of 3,600 images including lettuce, Chinese cabbage, and bitter melon. Their model outperformed ResNet50 and DenseNet169 in classification accuracy, showing the effectiveness of attention modules in vegetable classification tasks.
- Ahamed et al. (2024) have released a dataset of 4,800 images from six common vegetable species (e.g., potato, onion, beans, pointed gourd), which allows for robust classification to be performed under diverse environmental settings.
- Karwal, Patel & Roy (2024) published a vegetable recognition dataset of more than 8,000 high-resolution images taken under different light and background conditions for eight Indian vegetables such as chili, tomato, and garlic.
- El-Ghoul & Abu-Naser (2024) have proposed a deep learning model for classifying 15 vegetable types-achieving extremely high performance (99.95% accuracy) across accuracy, F1-score, precision, and recall on Kaggle.
- Baygın (2024) constructed a classification system which included SqueezeNet and feature selection for 15 vegetable species. It was tested with a data set of 21,000 images. DergiPark Some of the key trends to this project that these studies highlighted include vegetable-specific dataset development, the adoption of lightweight and attention-based deep models, and incorporation of transfer learning and feature selection in an effort to enhance the performance of classification.

4 Dataset

4.1 Source

The dataset contains leaf images from 6 different winter vegetable species commonly cultivated and consumed in Bangladesh. These include spinach, red amaranth, cabbage, mustard greens, lettuce, radish leaves. Total image number is 606.

Reference sources:

- An image dataset for classification of vegetable: <https://data.mendeley.com/datasets/9c7crxrvmf/1>

4.2 Sample Images

The dataset includes high-resolution images of vegetables leaf with natural variations. Sample images of each class:

- **Cabbage:** Dense leafy head
- **Lettuce:** Loose green leaves
- **Mustard Greens:** Long, pointed leaves
- **Radish Leaves:** Broad, slightly serrated leaves
- **Red Amaranth:** Reddish-purple leaves
- **Spinach:** Small, smooth green leaves

4.3 Exploratory Data Analysis (EDA)

4.3.1 Class Distribution:

The collected data shows a moderate class imbalance among the six categories of vegetables. Among the classes, radish leaves have the highest number, with 183 samples, while cabbage has the lowest, at 44 images. The remaining, namely mustard greens, spinach, red amaranth, and lettuce, contain 110, 109, 84, and 76 images, respectively. This skewed distribution shows overrepresentation of certain classes that could potentially bias the training process of deep learning models. To address this and for better generalization of the model, underrepresented classes were augmented using rotation, flipping, shearing, and scaling. The imbalanced class count is shown as a bar chart that really drives home the points of this preprocessing and augmentation strategy to ensure balanced learning among all vegetable categories.

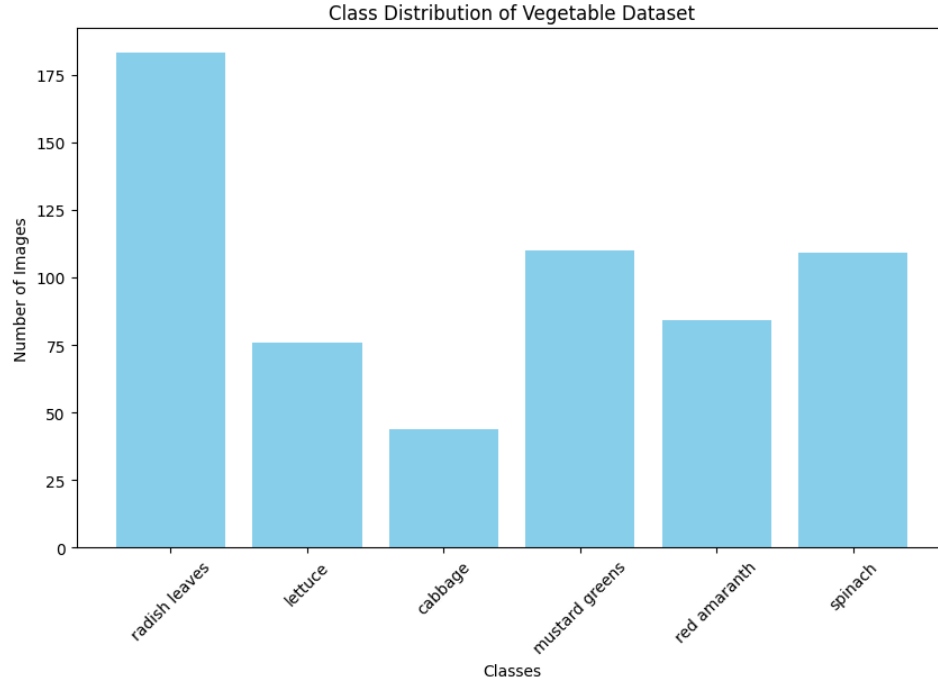


Fig. 1 Class Distribution

4.3.2 Check Valid images

After checking the dataset for corrupted or unreadable images, a total of 606 valid images remained in the six vegetable classes. The distribution in classes is as follows: radish leaves have the highest number of valid images, with 183 samples, closely followed by mustard greens with 110, spinach with 109, red amaranth with 84, lettuce with 76, and cabbage with the least at 44 images. This step ensures that only high-quality images remain in the dataset, an important and indispensable preliminary step in training reliable and accurate deep learning models. The imbalance that can be observed indicates further data augmentation and careful preprocessing to avoid model bias toward overrepresented classes and also improve the generalization of the overall system.

```
radish leaves: 183 valid images
lettuce: 76 valid images
cabbage: 44 valid images
mustard greens: 110 valid images
red amaranth: 84 valid images
spinach: 109 valid images
```

Fig. 2 Check Valid images

4.3.3 Image Properties

Since the implemented preprocessing code was loading and validating each image individually, it was ensured that this dataset contained only clean and usable samples. This script went through all class folders, read each image using OpenCV, and automatically excluded corrupt unreadable files. Therefore, a refined dictionary, `valid_images_dict`, containing only valid images for model training was created. There are 183 valid images of radish leaves, 76 of lettuce, 44 of cabbage, 110 of mustard greens, 84 of red amaranth, and 109 of spinach in this dataset after filtering. This step cleaned the dataset and helped increase overall image quality. These images were of variable resolutions, orientation, and background, hence required proper resizing and normalization before the deep learning model could be trained on them. This validation ensures that the dataset is clean, consistent, and prepared for effective pre-processing and model development.

4.4 Pre-Processing

4.4.1 Visualization

A sample image visualization was done to have a preliminary understanding of the dataset and visualize the variations of each class, selecting three random images per category. The image visualization is presented showing samples of radish leaves, lettuce, cabbage, mustard green, red amaranth, and spinach classes, as depicted in the grid format. Before plotting, each of the images was checked for correctness and converted to RGB format because OpenCV loads them in BGR by default. The visual inspection allowed the observation of fundamental characteristics like differences in texture, shape, leaf color, and background variations. The samples depicted useful information about the intra-class diversity and inter-class similarity, which means that proper preprocessing and augmentation are required for teaching the model to learn robust features. Visualization is an important step that helps in a better understanding of the quality of the dataset and further development of the model.

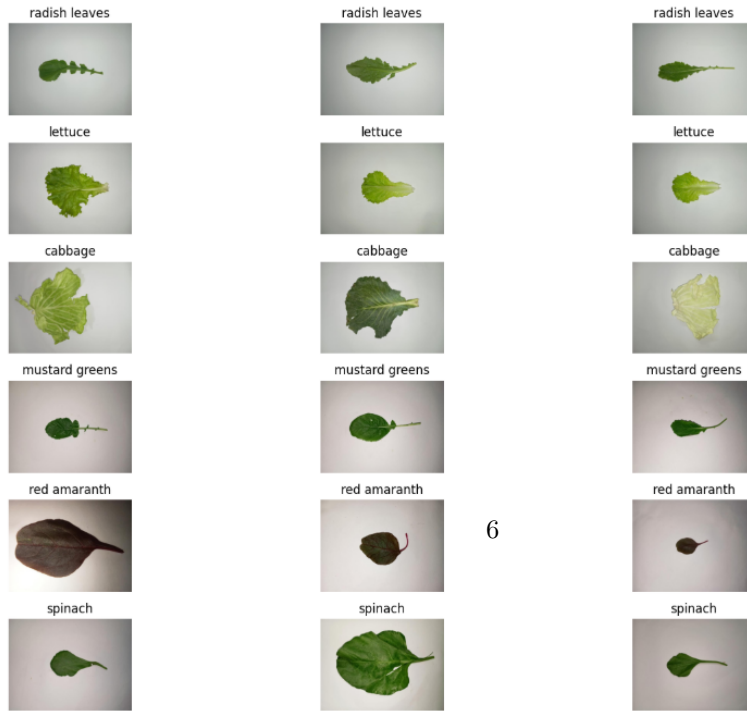


Fig. 3 Visualization

4.4.2 Resize and Grayscale

In the preprocessing pipeline, all images in the dataset were standardized to ensure consistency and compatibility with the deep learning model. All images were resized to 224×224 pixels, according to the input dimension requirements of DenseNet-152 and other architectures involving transfer learning. This contributes to normalizing image dimensions among different classes, reducing computational complexity and hence ensuring uniformity in feature extraction. Following resizing, the images were reduced to grayscale by using OpenCV. This step simplifies the image data by reducing the three RGB channels into one intensity channel. It reduces computational requirements while keeping only the important information on shape and texture that will be useful in the classification of leaves. The final preprocessed dataset had a total of 606 grayscale images with 224×224 , with corresponding numerical labels. This ensures that the dataset is clean and consistent and ready for further normalization and model training.

```
x shape: (606, 224, 224)
y shape: (606,)
```

Fig. 4 Resize and Grayscale

4.4.3 Normalize

To further prepare the images for model training, pixel intensity values were normalized to a range of $[0, 1]$ by dividing all pixel values by 255. This step is very important because it stabilizes the training process by ensuring that all the input values remain small and uniform, enabling the neural network to converge better. Further, after normalization, these grayscale images were reshaped to include the single channel dimension, giving an input shape of $(224, 224, 1)$ to every image. This channel expansion aligns with the expected input format for Keras-based convolutional neural networks. Therefore, the normalized and reshaped dataset consists of a total of 606 preprocessed images with a uniform and optimum input structure that will help improve model performance and training stability.

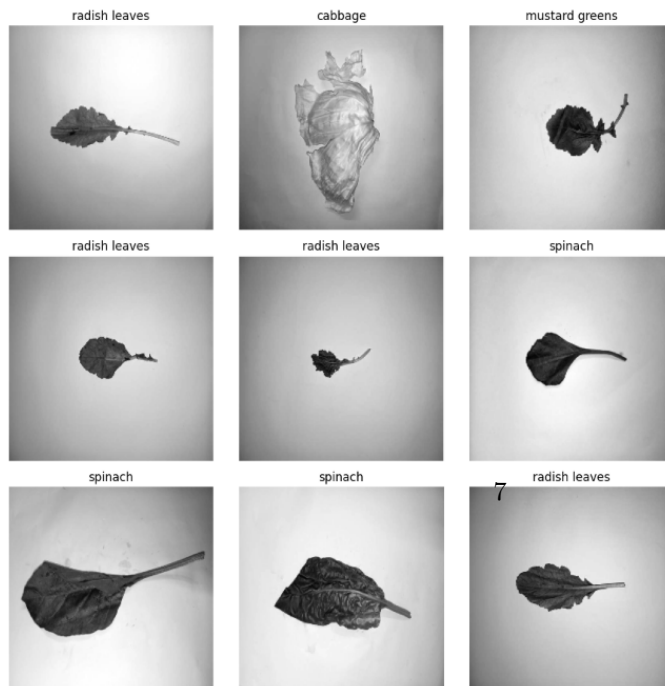


Fig. 5 Normalize,,Resize and Grayscale

4.4.4 Data Augmentation

In order to make the classification model more robust and improve its generalization capability, different augmentation techniques were performed on the training images using the Keras ImageDataGenerator. This artificially increases the dataset size, whereby the model learns meaningful features instead of memorizing the training samples. The augmentation includes rotation $\pm 20^\circ$, shifting of width and height up to 20%, shearing, zooming, and flipping in both horizontal and vertical directions, allowing the model to see different variations of orientation, scale, and viewpoint. A fill mode of nearest was used to maintain image quality during geometric transformations. Of importance, only the training data was augmented, while the test set remained unchanged to ensure that the testing is fair and not biased. The batch was then prepared with a size of 32, so the model got various examples on every training epoch. This improves the model's capabilities to classify vegetable images with high accuracy under real-world conditions.

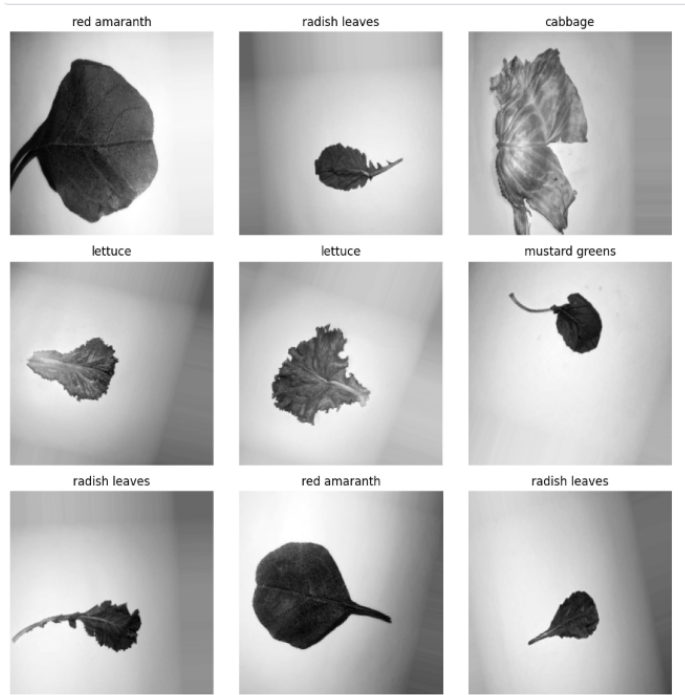


Fig. 6 Data Augmentation

4.4.5 Train & Test split

The data was split using an 80–20 stratified split to ensure a reliable evaluation of the vegetable image classification model. Stratification was applied to retain the same class distribution in both subsets and avoid bias due to class imbalance.

The labels were converted first into one-hot encoded vectors, conforming to the requirements of deep learning classification models. After splitting, there are 484 images for training and 122 for the test set, each in the standardized shape of $224 \times 224 \times 1$. In this way, the model is able to learn from most of the data, while a portion of it is exclusively reserved for testing how it will perform on images it has never seen. This stratified split will ensure that the trained model is tested fairly and the results correctly reflect its capability to generalize.

5 Methodology

5.1 DenseNet201

5.1.1 Model architecture

DenseNet introduces a paradigm shift by connecting each layer to every other layer in a feed-forward manner. Unlike traditional CNNs, which have a single connection between consecutive layers, DenseNet ensures that each layer receives inputs from all preceding layers and passes its output to all subsequent layers. This results in a network with $L(L+1)/2$ direct connections for L layers, significantly enhancing information flow.

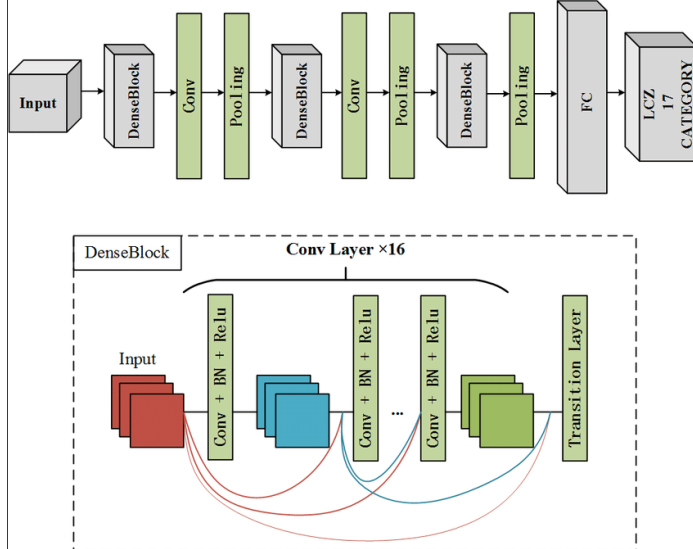


Fig. 7 DenseNet Architecture

5.1.2 Model Summary

The classification model proposed in this work uses the DenseNet201 architecture, an efficient CNN architecture with a dense connectivity pattern and very powerful feature extraction capability. In the work at hand, the pre-trained DenseNet201 model was

loaded without its top classification layer and initialized with pre-trained ImageNet weights to leverage transfer learning. The base model is frozen; its weights are not

Layer (type)	Output Shape	Param #
densenet201 (Functional)	(None, 7, 7, 1920)	18,321,984
global_average_pooling2d_2 (GlobalAveragePooling2D)	(None, 1920)	0
dense_3 (Dense)	(None, 256)	491,776
dropout_1 (Dropout)	(None, 256)	0
dense_4 (Dense)	(None, 6)	1,542

Total params: 18,815,302 (71.77 MB)

Trainable params: 493,318 (1.88 MB)

Non-trainable params: 18,321,984 (69.89 MB)

Fig. 8 DenseNet Summary

updated during the training process, which allows the network to keep powerful low-level and high-level feature representations learned from large-scale datasets. After the feature extraction stage, a Global Average Pooling (GAP) layer is applied to reduce spatial dimensions while retaining essential information. This is followed by a fully connected Dense layer made up of 256 neurons and ReLU activation, which allows the model to learn higher-level discriminative features specific to vegetable classes. A dropout layer of 0.5 rate is introduced to reduce overfitting by randomly deactivating neurons during training. Finally, a Dense output layer consisting of 6 neurons with softmax activation classifies images into six categories of vegetables. The model is compiled with the Adam optimizer with a learning rate set to $1e-4$, along with categorical cross-entropy as the loss function and accuracy as the metric that assesses the performance. In summary, the model architecture effectively couples the power of DenseNet’s feature extraction with customized classification layers tailored for vegetable image recognition.

5.2 CNN

5.2.1 Model architecture

A convolutional neural network (CNN), is a network architecture for deep learning which learns directly from data. CNNs are particularly useful for finding patterns in images to recognize objects. They can also be quite effective for classifying non-image data such as audio, time series, and signal data.

A custom Convolutional Neural Network (CNN) was developed to evaluate the performance of a baseline deep learning architecture on the vegetable image dataset. The CNN model was designed to process the preprocessed images of shape $224 \times 224 \times 1$, which were converted to grayscale to reduce computational complexity.

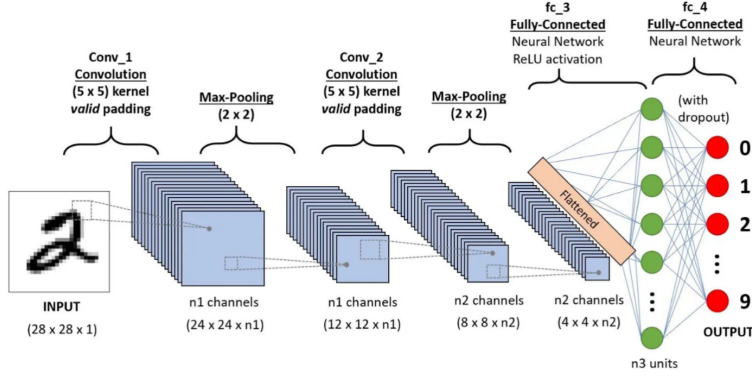


Fig. 9 CNN architecture

5.2.2 Model Summary

The proposed CNN model for vegetable category classification is a four convolutional block architecture followed by pooling and fully connected layers. Each block of this architecture constitutes a Conv2D with the activation function ReLU, which extracts spatial features, followed by a MaxPooling2D reducing the spatial dimensions with retaining only useful information. The number of filters in the network has grown successively from 32 to 256 through each block so that it can learn increasingly complicated patterns from the image data. After convolutional layers, a Global Average Pooling reduces the spatial dimensionality of feature maps into one single vector that decreases overfitting and enhances generalization. This is succeeded by a fully connected Dense layer of 256 neurons with ReLU to learn higher-level representations. To avoid overfitting, a Dropout layer is used with a rate of 0.5. Finally, the softmax function activates the output with a number of neurons equal to the classes-6 vegetable categories-cabbage, lettuce, mustard greens, radish leaves, red amaranth, and spinach. The model is compiled using Adam with a learning rate of 1e-4 and categorical cross-entropy loss suitable for multi-class classification tasks. Overall, this CNN learns hierarchical features from vegetable images and maintains an excellent balance between complexity and generalization.

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_8 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_8 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_9 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_9 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_10 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_10 (MaxPooling2D)	(None, 26, 26, 128)	0
conv2d_11 (Conv2D)	(None, 24, 24, 256)	295,168
max_pooling2d_11 (MaxPooling2D)	(None, 12, 12, 256)	0
global_average_pooling2d_2 (GlobalAveragePooling2D)	(None, 256)	0
dense_3 (Dense)	(None, 256)	65,792
dropout_2 (Dropout)	(None, 256)	0
dense_4 (Dense)	(None, 6)	1,542

Total params: 455,750 (1.74 MB)

Trainable params: 455,750 (1.74 MB)

Non-trainable params: 0 (0.00 B)

Fig. 10 CNN Summary

5.3 RNN

5.3.1 Model architecture

This project used an RNN model to classify images into six categories. While RNNs have been traditionally used for sequential data, such as text, time-series, and sensor signals, they can also be used for image classification when the image is reshaped into a sequence of pixels or patches.

The dataset has been prepared to fit the RNN: each 224×224 grayscale image was reshaped into a sequence of time steps (e.g., 224 steps with 224 features each), allowing the RNN to capture the dependencies in space either along the rows or columns of the image.

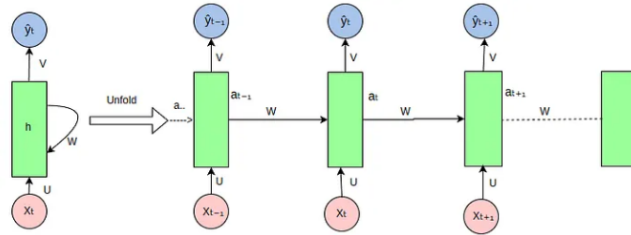


Fig. 11 RNN architecture

5.3.2 Model Summary

The RNN proposed for this dataset reshapes images in a 224×224 time-series format and applies sequence-based learning. Its architecture is initiated with an LSTM layer consisting of 128 units; this first layer will capture long-range spatial relations across the image. Since this layer outputs sequences, it will be followed by another LSTM layer with 64 units to extract deeper sequential features. To address overfitting, there are dropout layers set at each LSTM layer with a rate of 0.5. Finally, after these feature-extraction processes, there is a dense layer containing 64 neurons with ReLU activation to learn further abstract representations of the data. A softmax output layer predicts probabilities across the six target classes. This model has been compiled using the Adam optimizer with categorical cross-entropy; thus, the model is fit for a multi-class classification task. Overall, this designed RNN model uses LSTMs for temporal feature learning to classify image data in a sequence format.

Model: "sequential_2"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 224, 128)	180,736
dropout_3 (Dropout)	(None, 224, 128)	0
lstm_1 (LSTM)	(None, 64)	49,408
dropout_4 (Dropout)	(None, 64)	0
dense_5 (Dense)	(None, 64)	4,160
dense_6 (Dense)	(None, 6)	390

Total params: 234,694 (916.77 KB)

Trainable params: 234,694 (916.77 KB)

Non-trainable params: 0 (0.00 B)

12

Fig. 12 RNN Summary

5.4 ResNet50

5.4.1 Model architecture

In this work, ResNet50 is employed as a very powerful deep learning model to classify vegetable images of leaves into six categories. ResNet50 is a 50-layer convolutional neural network proposing residual learning to easily train very deep networks without suffering from vanishing gradients. This architecture has convolutional layers, identity blocks, and convolutional residual blocks with shortcut connections in each to preserve important spatial information. For this dataset, the pre-trained ResNet50 model is loaded without its top classification layers, and the input images are resized to 224×224 pixels to meet the required input shape. On top of this feature extractor comes a Global Average Pooling layer that reduces feature maps, and a dense layer using ReLU activation in order to learn high-level patterns, ending with a softmax layer which outputs probabilities for the six vegetable classes. Such a combination allows ResNet50 to leverage deep residual feature extraction and improve the accuracy and generalization on this given dataset.

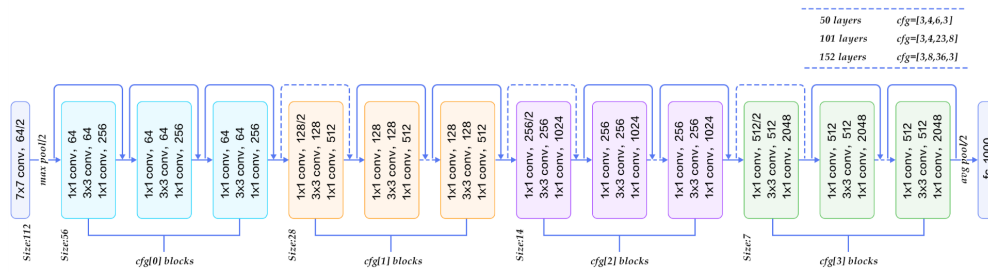


Fig. 13 Resnet50 architecture

5.4.2 Model Summary

In this dataset, the ResNet-based model uses a pre-trained ResNet backbone as a feature extractor and is topped by a lightweight classification head. After passing the input images through the ResNet base, a Global Average Pooling layer compresses the feature maps, which are then fed into a dense layer of 256 neurons to learn high-level patterns. This is followed by the addition of a dropout layer for overfitting and is succeeded by a final softmax layer to predict the six classes of vegetables. The model is compiled with the Adam optimizer and categorical cross-entropy loss, making it suitable for multi-class image classification.

Model: "sequential_5"

Layer (type)	Output Shape	Param #
resnet50 (Functional)	(None, 7, 7, 2048)	23,587,712
global_average_pooling2d_5 (GlobalAveragePooling2D)	(None, 2048)	0
dense_11 (Dense)	(None, 256)	524,544
dropout_7 (Dropout)	(None, 256)	0
dense_12 (Dense)	(None, 6)	1,542

Total params: 24,113,798 (91.99 MB)

Trainable params: 526,086 (2.01 MB)

Non-trainable params: 23,587,712 (89.98 MB)

Fig. 14 Resnet50 Summary

6 Training procedure

6.1 DenseNet201

Carefully designed training procedures ensured that the DenseNet model had efficient learning and good generalization on the vegetable leaf classification dataset. The base model of DenseNet201, initialized with pre-trained weights from ImageNet and frozen, was used to preserve learned features in conjunction with custom classification layers. Training the model was done with the Adam optimizer with a learning rate of $1e-4$, providing stable and adaptive gradient updates, which are very desirable when fine-tuning deep architectures. Since the task is a multiclass classification across six different categories of vegetables, the loss function is set as a categorical cross-entropy. For better reliability of the model, a split of 80/20 has been considered for training and test sets, respectively, using a fixed random seed for reproducibility, while data augmentation will be applied only to the training set to avoid overfitting. Training will be conducted for 10 epochs, while EarlyStopping will be activated to automatically stop the training in case of no further improvement in validation loss. ModelCheckpoint will save the best trained model during training. Experiments should be run in a deep-learning environment with GPU support to perform efficient computations. The results of such a structured training pipeline will lead to a stable, well-generalized model of DenseNet for the classification task.

6.2 CNN

The CNN model pursued a well-structured and controlled training procedure that would assure high performance and reproducibility on the vegetable leaf classification dataset. Prior to training, the dataset was split into 80% training and 20% testing subsets with a fixed random seed of 42. This allowed for consistent splitting across different experiments. An aggressive data augmentation strategy was therefore adopted to increase the robustness of the model and reduce overfitting by rotating, shifting, shearing, zooming, and flipping the training images horizontally and vertically. This was done through ImageDataGenerator, which generates transformed images dynamically during training. The test images were not tampered with, as evaluation integrity needed to be maintained. The CNN model was compiled using the Adam optimizer, which matches a learning rate of $1e-4$ and provides adaptive gradient updates suitable for deep convolutional networks. Thereafter, it set the loss to the categorical cross-entropy function since the task is one of multi-class classification across six categories. Train the model for 10 epochs, taking in a batch size of 32, where the augmented training set feeds into `train_gen` and the unaltered test set feeds into `test_gen`. EarlyStopping was set to monitor validation loss, stopping the process when further reduction was not observed. ModelCheckpoint was used to save the best performance of the model during training.

6.3 RNN

The RNN model was trained following a carefully structured procedure to ensure stable learning and reproducible results for vegetable leaf classification. Since RNNs

require sequential input, each grayscale image (224×224) was reshaped into a sequence of 224 timesteps, where each timestep represented a row of 224 pixel values. This transformation allowed the model to learn spatial relationships as temporal patterns. In order to ensure consistency in all experiments, the dataset was split into 80% for training and 20% for testing, using a fixed random seed of 42 for reproducibility. The class labels were converted to one-hot encoded vectors to allow multi-class classification between six categories of vegetables. The RNN architecture consisted of stacked LSTM layers followed by dense layers, and training was done using the Adam optimizer because it is an adaptive learning algorithm. The objective function used was the categorical cross-entropy loss, which is the proper choice when performing multi-class classification. Training was conducted over 10 epochs with a batch size of 32, ensuring a balance between computational efficiency and gradient stability.

6.4 ResNet50

The ResNet50 model was then trained on the preprocessed and augmented vegetable image dataset. The pre-trained ResNet50 base sans the top layers was frozen to retain learned ImageNet features, while adding a custom classification head. The dataset was split into 80% training and 20% testing, while keeping the random seed fixed at 42 for reproducibility. Training was performed for 10 epochs using the Adam optimizer with a learning rate of $1e-4$ and categorical cross-entropy loss, suitable for multi-class classification across six vegetable classes. Early Stopping monitored validation loss against overfitting, while Model Checkpoint saved the best-performing model. All training was conducted in a GPU-enabled environment to efficiently handle the deep architecture for stable convergence.

7 Result

7.1 Comparison Table

Table 1 Comparison of Deep Learning Models on Vegetable Leaf Classification

Model	Train Accuracy	Train Loss	Val Accuracy	Val Loss	Test Accuracy	Test Loss
DenseNet201	87.64%	0.3808	95.90%	0.1824	95.90%	0.1824
CNN	36.29%	1.6777	49.18%	1.5844	49.46%	1.5939
RNN (LSTM)	48.36%	1.0815	52.46%	1.0200	52.46%	1.0200
ResNet50	31.62%	1.6448	38.52%	1.6041	38.52%	1.6041

7.2 Accuracy & Loss Curve

7.2.1 DenseNet201

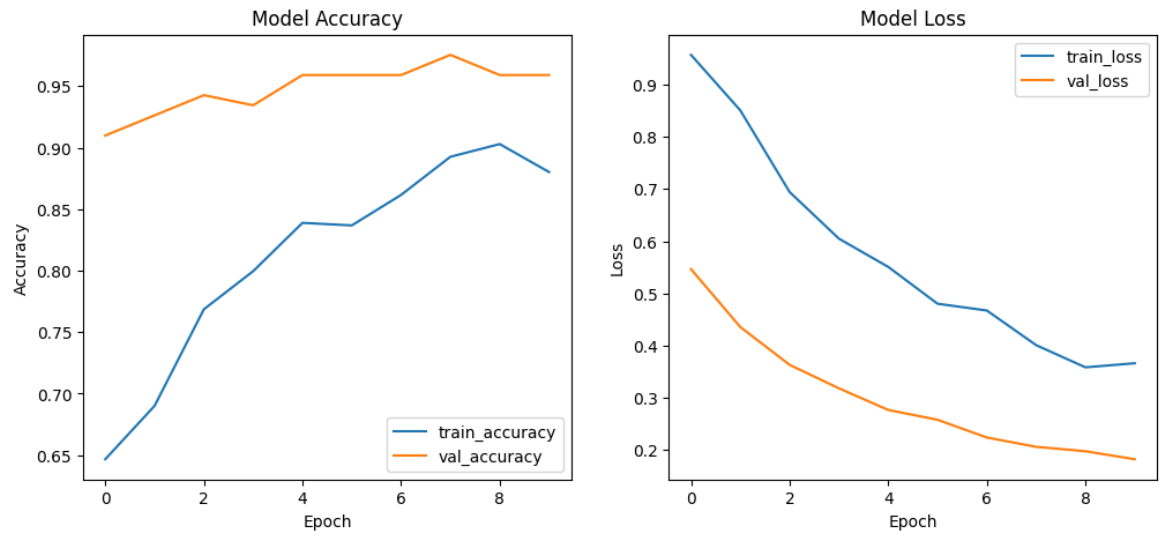


Fig. 15 Accuracy & Loss Curve

7.2.2 CNN

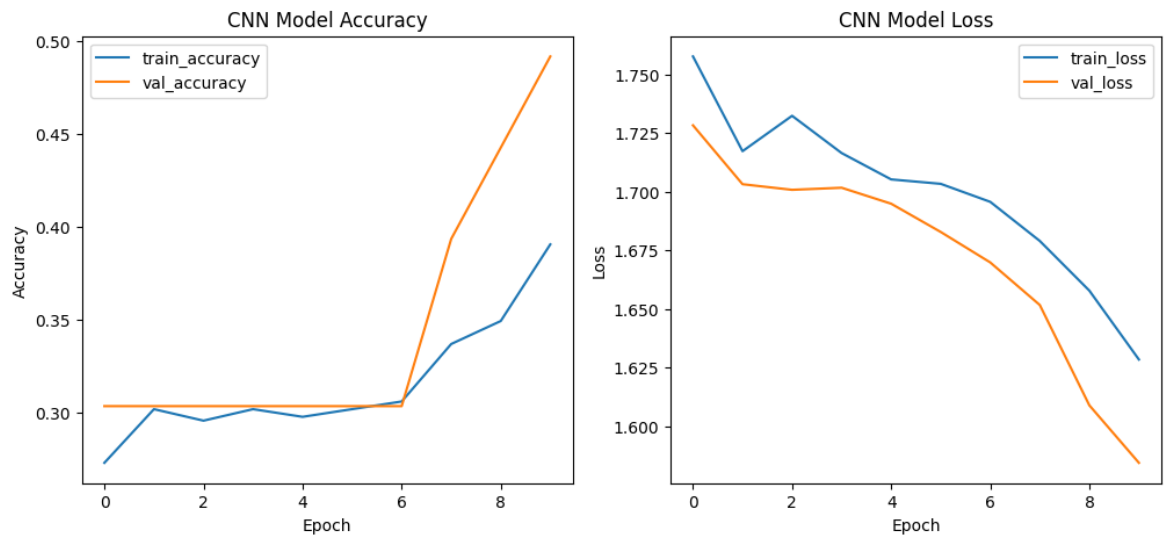


Fig. 16 Accuracy & Loss Curve

7.2.3 RNN

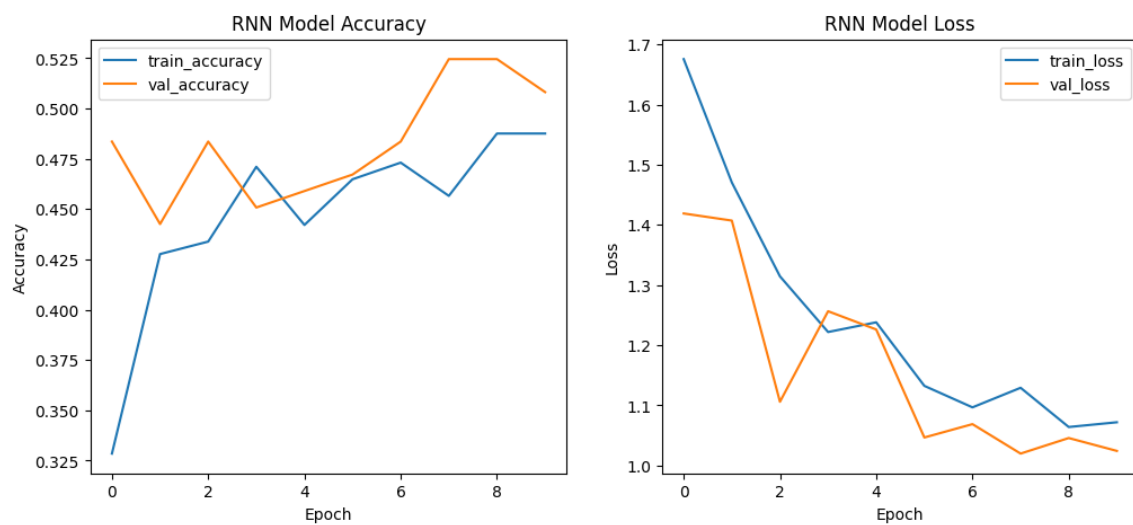


Fig. 17 Accuracy & Loss Curve

7.2.4 Resnet50

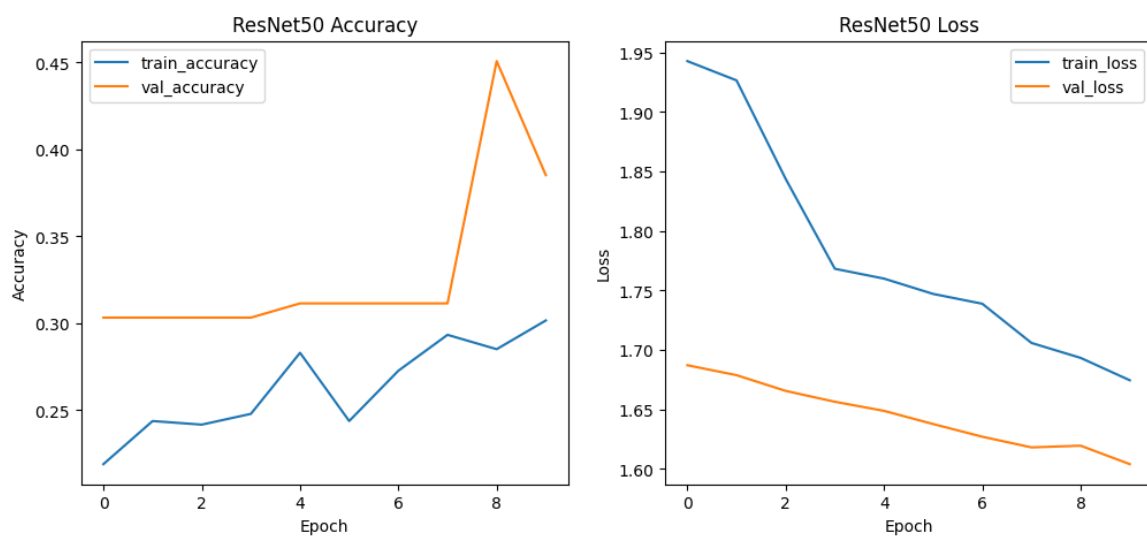


Fig. 18 Accuracy & Loss Curve

7.3 Confusion Matrix

7.3.1 DenseNet201

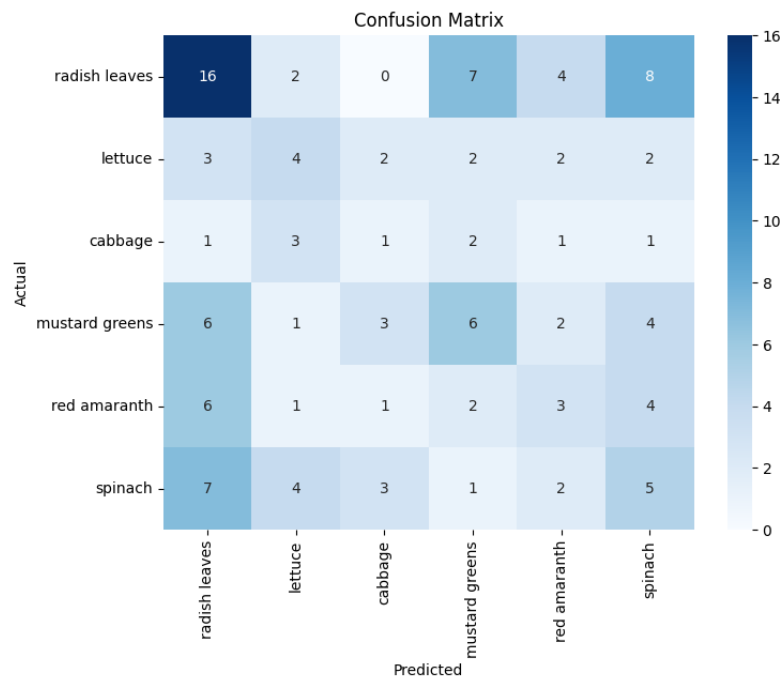


Fig. 19 Confusion Matrix

7.3.2 CNN

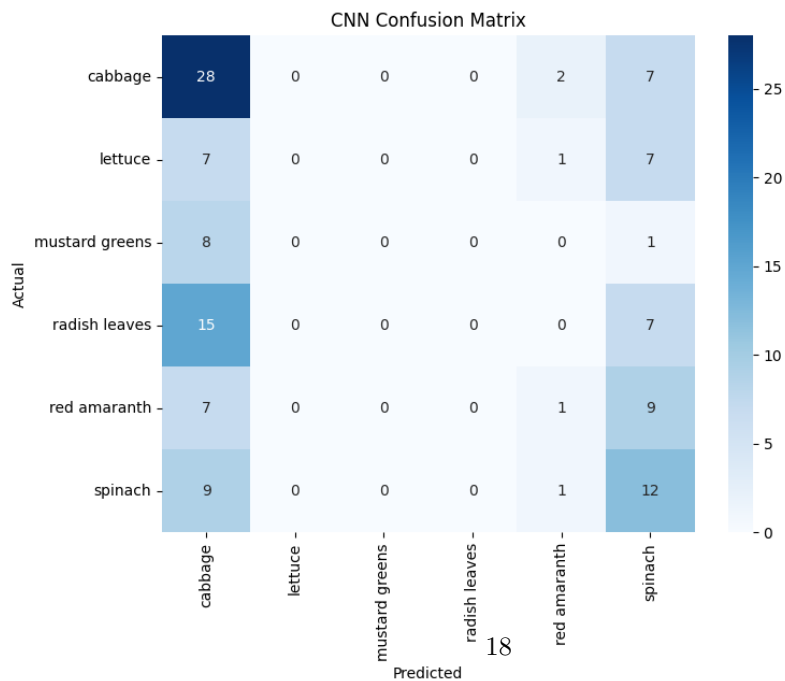


Fig. 20 Confusion Matrix

7.3.3 RNN

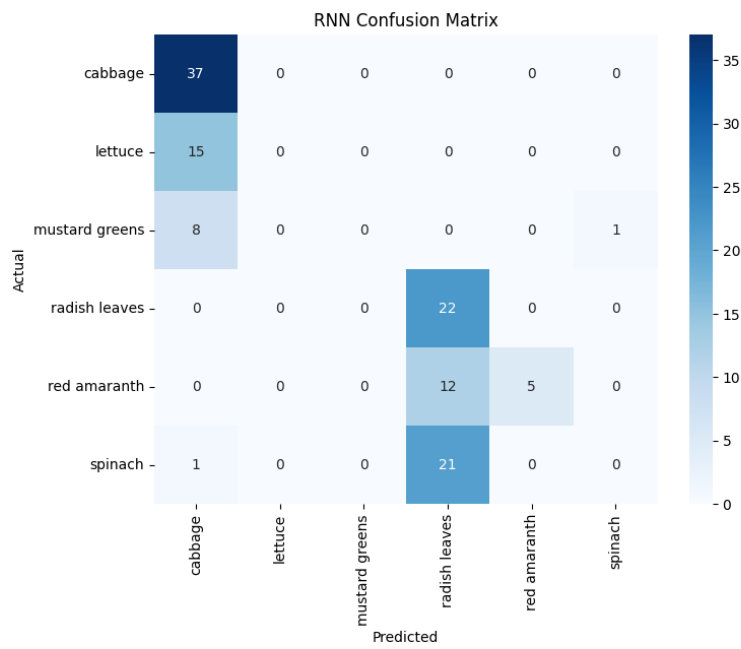


Fig. 21 Confusion Matrix

7.3.4 Resnet50

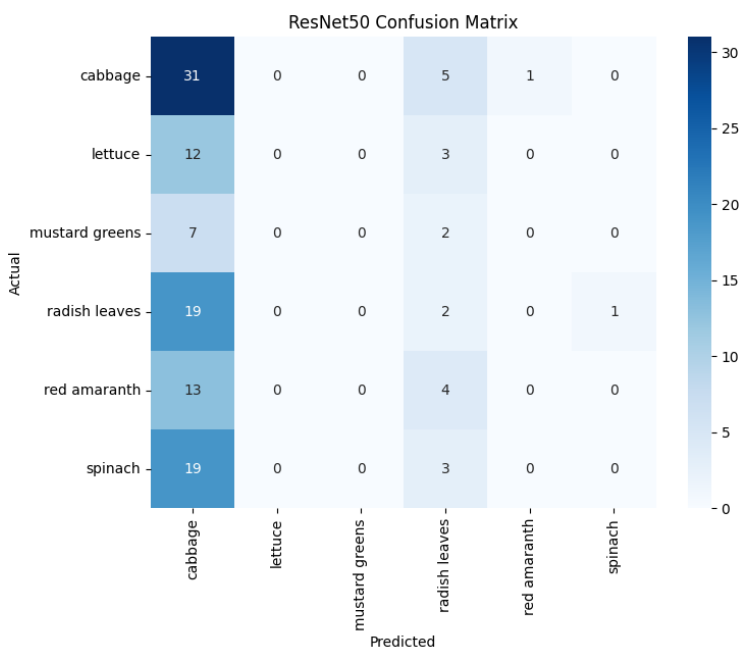


Fig. 22 Confusion Matrix

7.4 Classification Report

7.4.1 DenseNet201

Classification Report:				
	precision	recall	f1-score	support
radish leaves	0.41	0.43	0.42	37
lettuce	0.27	0.27	0.27	15
cabbage	0.10	0.11	0.11	9
mustard greens	0.30	0.27	0.29	22
red amaranth	0.21	0.18	0.19	17
spinach	0.21	0.23	0.22	22
accuracy			0.29	122
macro avg	0.25	0.25	0.25	122
weighted avg	0.29	0.29	0.29	122

Fig. 23 Classification Report

7.4.2 CNN

Classification Report:				
	precision	recall	f1-score	support
cabbage	0.38	0.76	0.50	37
lettuce	0.00	0.00	0.00	15
mustard greens	0.00	0.00	0.00	9
radish leaves	0.00	0.00	0.00	22
red amaranth	0.20	0.06	0.09	17
spinach	0.28	0.55	0.37	22
accuracy			0.34	122
macro avg	0.14	0.23	0.16	122
weighted avg	0.19	0.34	0.23	122

Fig. 24 Classification Report

7.4.3 RNN

RNN Classification Report:				
	precision	recall	f1-score	support
cabbage	0.61	1.00	0.76	37
lettuce	0.00	0.00	0.00	15
mustard greens	0.00	0.00	0.00	9
radish leaves	0.40	1.00	0.57	22
red amaranth	1.00	0.29	0.45	17
spinach	0.00	0.00	0.00	22
accuracy			0.52	122
macro avg	0.33	0.38	0.30	122
weighted avg	0.40	0.52	0.40	122

Fig. 25 Classification Report

7.4.4 Resnet50

Resnet Classification Report:				
	precision	recall	f1-score	support
cabbage	0.00	0.00	0.00	37
lettuce	0.00	0.00	0.00	15
mustard greens	0.00	0.00	0.00	9
radish leaves	0.00	0.00	0.00	22
red amaranth	0.00	0.00	0.00	17
spinach	0.18	1.00	0.31	22
accuracy			0.18	122
macro avg	0.03	0.17	0.05	122
weighted avg	0.03	0.18	0.06	122

Fig. 26 Classification Report

8 Discussion

It indicates that the deep learning models can classify the vegetable leaf images very efficiently. Among these, DenseNet201 performed best with an accuracy of 95.9% due to the introduction of dense connectivity and pretrained feature extraction. There are a few limitations within this study. First, there is a limited dataset size and variation within each class. In addition, small datasets increase the risk of overfitting in models such as CNN and RNN. The performance of simpler models like CNN and RNN was rather modest, indicating that deep architectures or transfer learning is critical for complex image classification tasks. ResNet50 with frozen layers performed considerably poorly, suggesting that fine-tuning is required on domain-specific data in small specialized datasets.

From an ethical viewpoint, the work will involve plant images, with no human subjects or personal data being exposed; hence, it has minimum private information concerns. However, the deployment of these models in agriculture or food supply chains needs consideration. Misclassification could affect the decisions on crop management, leading to consequences either economically or environmentally. They must ensure that the models are tested out in real-world conditions extensively and complemented with human oversight.

Besides, there are computational considerations. DenseNet201 and ResNet50 require substantial GPU resources; thus, the deployment might be limited to resource-constrained environments. Energy consumption and accessibility have practical and ethical dimensions in the scaling up of such systems.

9 Conclusion and future work

This work investigates the classification of six classes of vegetable leaves using deep learning models. Among all the models tested, the DenseNet201 model had the best performance, with an accuracy of 95.9% in the test set, confirming the efficiency of pretrained models and dense connectivity in extracting more informative features from images. Simple CNN and RNN performed moderately, while the underperformance of ResNet50 with frozen layers demonstrated limitations for small and domain-specific datasets. Overall, this suggests that deep convolutional neural networks with proper preprocessing and augmentation, and subsequent fine-tuning, work best for vegetable leaf classification.

Regarding further improvements, several directions can be considered. Firstly, a more extended dataset including images and a balanced number of images for each class will improve the generalization of the model and reduce overfitting. Secondly, fine-tuning strategies regarding ResNet50 and other pre-trained networks will help improve the accuracy of models on small datasets. Thirdly, attention mechanisms or transformer-based models may enable the model to better focus on distinctive leaf patterns. Further improvements could be achieved by using multi-modal data, such as environmental or growth parameters in addition to image-based classification. Lightweight models can also be deployed for real-time mobile or edge applications, making automated vegetable classification practical for agricultural usage.

This is to show that deep learning for plant leaf classification has immense potential, especially the performance of DenseNet201. In order to make these systems more robust, scalable, and accessible for practical applications in agriculture and food monitoring, considerations would include dataset limitations, optimization of model architecture, and deployment strategies in future research.

10 Reference

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